

Geothermal power and reservoir prediction

Machine Learning Assignment 1 | Roll number: **BT2024186**

1. Data and approach

The two personalised problems were modelled separately. Each has 1,000 labelled training rows and 1,000 unlabelled test rows. Var1 supplies six operational inputs; var2 supplies three spatial inputs. All values are finite, no values are missing, and inputs are bounded between -1 and 1. The original preprocessing was retained.

Each model uses all monomials whose total degree is at most d , including feature interactions. The fitted function is $\hat{y} = b + \sum(\beta_j * \phi_j(x))$, where ϕ_j is a polynomial term. PolynomialFeatures excludes a constant column because the regression fits an intercept. Expanded columns are standardized within the training pipeline to improve numerical stability.

2. Selection and independent evaluation

A reproducible 80/20 split (seed 42) reserves 200 rows per problem for holdout evaluation. On the remaining 800 rows, shuffled five-fold cross-validation compares degrees, feature sets, and ordinary least squares, ridge, and lasso regression. The scaler is fitted inside each fold. Candidate selection uses only cross-validation MSE; holdout scores and test inputs do not select hyperparameters. Final models are refitted on all 1,000 labelled rows.

The one-standard-error rule selects the polynomial with the fewest expanded terms among candidates with MSE no greater than the minimum CV MSE plus the best candidate's fold standard deviation divided by $\sqrt{5}$. Within that term count, the lowest MSE wins. This is a practical complexity heuristic, rather than a formal confidence interval.

Problem	Degree / inputs	Holdout MSE	Holdout R2
var1	5 / 6 inputs	0.383402	0.964099
var2	10 / 3 inputs	0.273390	0.994922

These are validation results on training data, not scores against the hidden test ground truth.

Steam turbine optimization

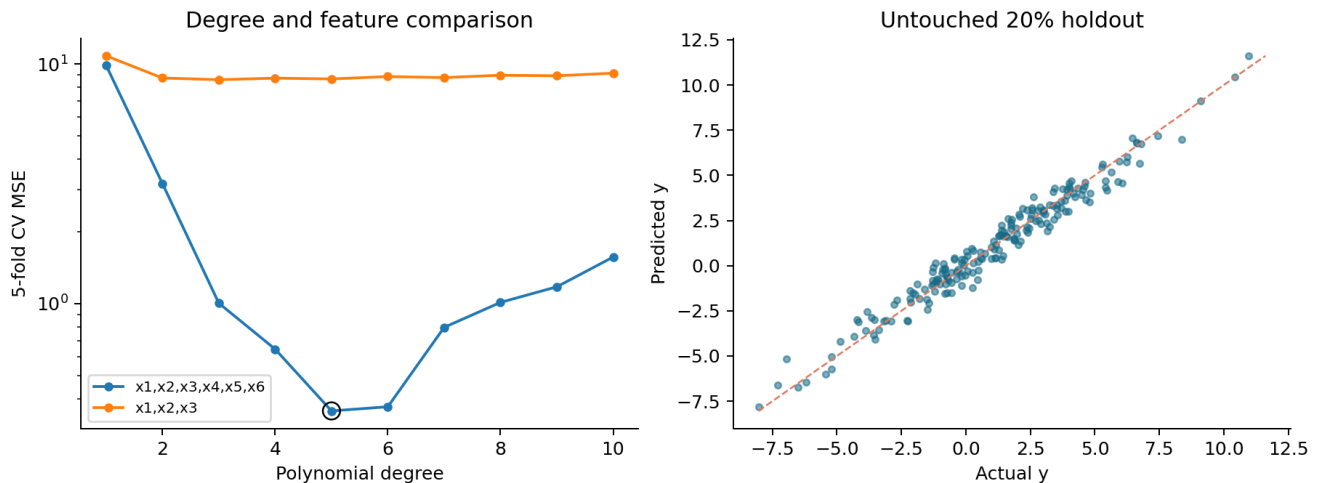
Selected: **degree 5**, inputs **x1, x2, x3, x4, x5, x6**, Lasso with alpha = **0.01**.

3. Search and rationale

The search compared x1-x3 at degrees 1-10 with all six inputs at degrees 1-10. OLS and SVD ridge were evaluated through degree 6 for all inputs; higher degrees used iterative ridge with alpha 1, 10, or 100 to handle the large design matrices. Lasso was evaluated at degrees 2-6. The reduced feature set was tested explicitly.

A total of 149 candidates were evaluated. The chosen model has 461 expanded terms excluding the intercept, of which 115 have nonzero fitted coefficients. Its CV MSE is 0.356319 with fold standard deviation 0.042444; the minimum CV MSE is 0.356319. The one-standard-error threshold is 0.375300.

Inputs / degree	Best regression	Alpha	CV MSE
All inputs / 1	Ridge	100	9.864050
All inputs / 4	Lasso	0.01	0.645032
All inputs / 5	Lasso	0.01	0.356319
All inputs / 6	Lasso	0.01	0.370923
Reduced inputs / 3	Ridge	10	8.599024



Left: minimum CV MSE across tested regressors at each degree; circle marks the selected model. Right: holdout predictions against observations, with a perfect-prediction reference line.

Holdout MSE = **0.383402**; RMSE = **0.619195**; R2 = **0.964099**. A mean-only baseline has holdout MSE 10.745735. The final full-data training MSE is 0.254201; it is a fitting diagnostic and does not replace validation.

Thermal reservoir mapping

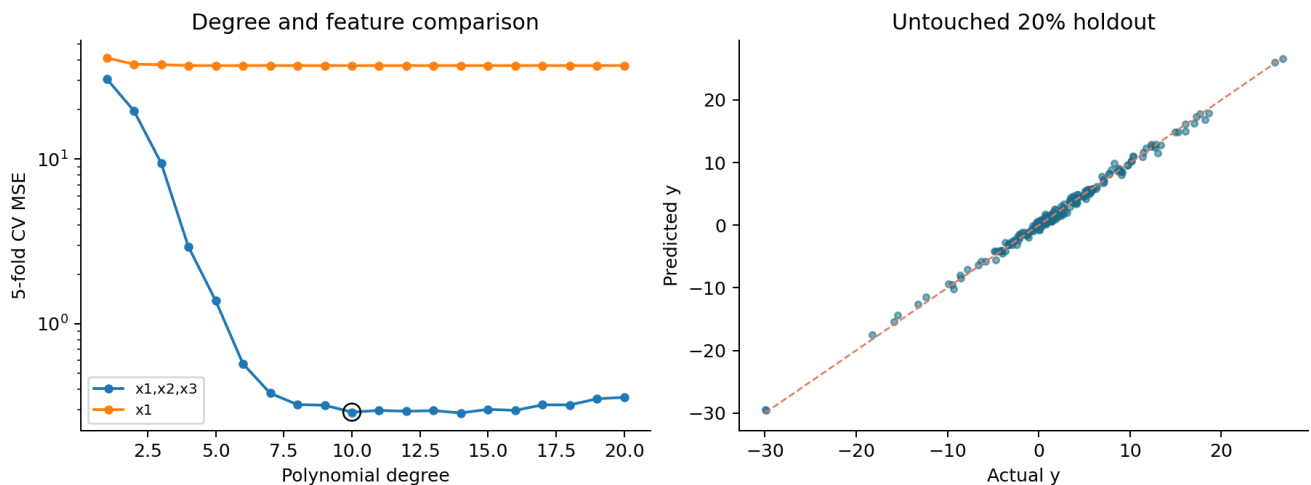
Selected: **degree 10**, inputs **x1, x2, x3**, **Lasso** with $\alpha = 0.001$.

4. Search and rationale

The search compared x1 alone with all three inputs at every degree from 1 to 20. OLS and SVD ridge were evaluated for both feature sets; lasso was evaluated on all three inputs at degrees 4-12. The single-feature alternative was tested explicitly.

A total of 325 candidates were evaluated. The chosen model has 285 expanded terms excluding the intercept, of which 126 have nonzero fitted coefficients. Its CV MSE is 0.290095 with fold standard deviation 0.076045; the minimum CV MSE is 0.286985. The one-standard-error threshold is 0.312791.

Inputs / degree	Best regression	Alpha	CV MSE
All inputs / 1	Ridge	10	30.834460
All inputs / 9	Ridge	1	0.319690
All inputs / 10	Lasso	0.001	0.290095
All inputs / 11	Lasso	0.001	0.297265
Reduced inputs / 9	Ridge	100	37.085596



Left: minimum CV MSE across tested regressors at each degree; circle marks the selected model. Right: holdout predictions against observations, with a perfect-prediction reference line.

Holdout MSE = **0.273390**; RMSE = **0.522867**; R² = **0.994922**. A mean-only baseline has holdout MSE 53.842100. The final full-data training MSE is 0.180220; it is a fitting diagnostic and does not replace validation.

Implementation and deliverables

5. Regularization and numerical choices

Ridge minimizes squared residual error plus alpha times the squared coefficient norm. Lasso instead penalizes the absolute coefficient norm and can remove unnecessary monomials. Both remain polynomial regression: prediction is a linear combination of polynomial terms. The intercept is unpenalized. Ridge alphas are 0.0001, 0.01, 0.1, 1, 10, and 100; lasso alphas are 0.001, 0.003, 0.01, 0.03, and 0.1. Large var1 expansions use a narrower ridge grid.

SVD ridge handles correlated polynomial columns; high-degree var1 ridge uses LSQR with tolerance 1e-8 and at most 10,000 iterations. Lasso uses tolerance 1e-5 and at most 50,000 iterations. Some high-degree lasso candidates reached the iteration limit; both selected models were checked to converge on every CV fold. BLAS threads are limited to one for repeatability and to avoid oversubscription.

6. Code and reproduction

Measured environment: Python 3.13.1; NumPy 2.3.5; pandas 2.3.3; scikit-learn 1.8.0. No GPU is required. Run **python train.py** from the repository root to repeat selection, validation, final fitting, and prediction. Run **python build_report.py** after training to regenerate this report.

For inference without retraining, run **python predict.py --model artifacts/var1_model.joblib --input BT2024186/BT2024186_test_var1.csv --output predictions.csv**. Use the analogous var2 paths for the second problem. README.md contains Windows commands and dependency instructions.

Deliverable	Contents
BT2024186_pred_var1.csv	1,000 predictions; a single y column; original test row order
BT2024186_pred_var2.csv	1,000 predictions; a single y column; original test row order
BT2024186_report.pdf	This four-page report
artifacts/	Saved models, coefficients, fold scores, holdout predictions, metrics and plots

7. Validation and limitations

The prediction files follow the provided sample submission: header y, no index and no added feature columns. Each row corresponds directly to the same row in its original test CSV. Saved-model predictions are checked against the submitted files, and polynomial coefficients are exported in the original input basis for inspection.

Hidden test targets are unavailable, so test MSE and R2 cannot be computed locally. The reported holdout uses one reproducible random split and has sampling uncertainty. Degree and feature choices are justified by cross-validation on these personalised datasets; no external data, non-polynomial predictor, test labels, or target-based preprocessing is used.